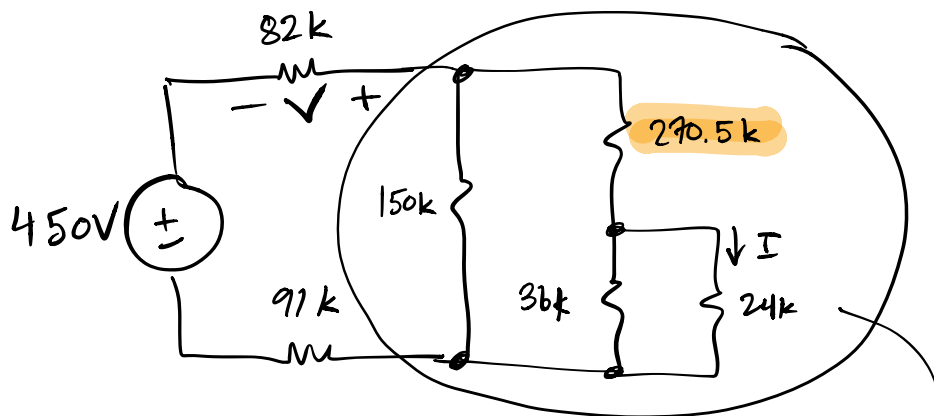
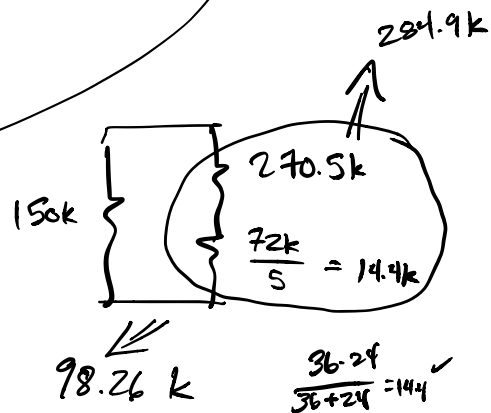
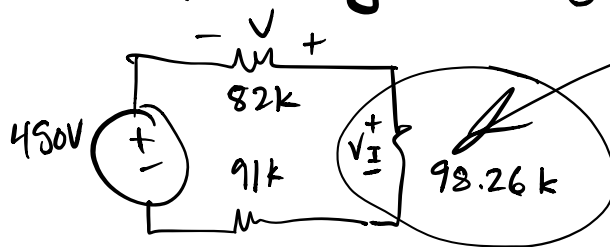


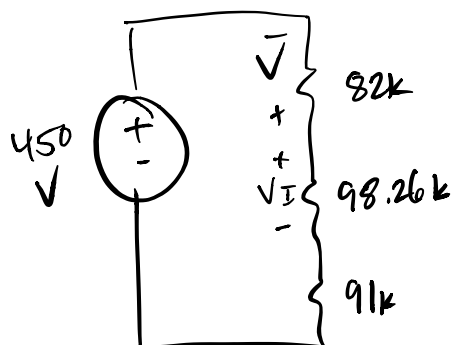


.. let's find V first

.. temporarily simplify further



.. use voltage division to find V and V_I .



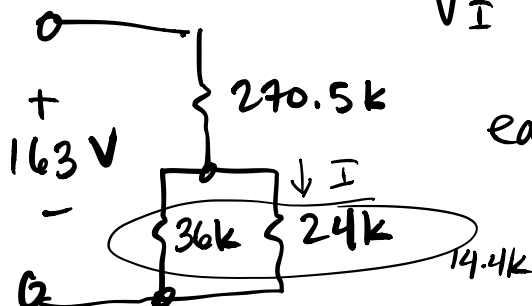
$$-V = 450 \left(\frac{82k}{82k + 98.26k + 91k} \right)$$

$$= 136V$$

$$V = -136V$$

$$V_I = 450 \left(\frac{98.26}{98.26 + 82 + 91} \right) = 163V$$

easiest: another voltage divider



$$163 \left(\frac{14.4k}{14.4k + 270.5k} \right) = 8.239V$$

$$I = \frac{8.239V}{24k} = 0.3433mA$$

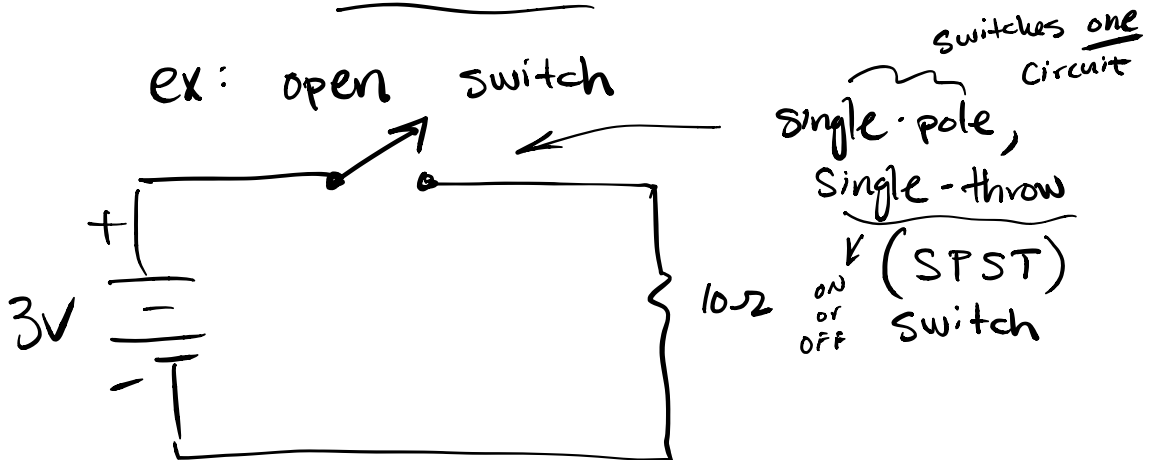
or
343.3
μA

Open and Short Circuits

open circuit: equivalent to infinite resistance

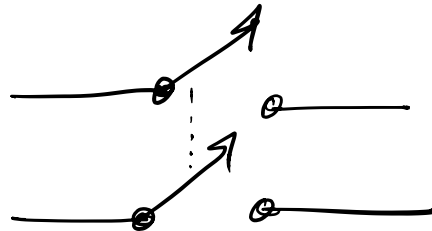
→ interruption of a closed circuit

ex: open switch



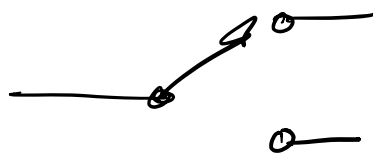
-- switch is shown open (OFF)

-- no current can flow because resistance is infinite



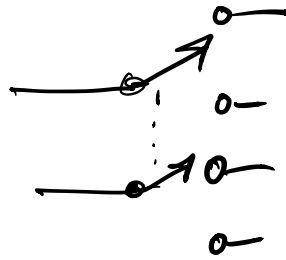
double-pole,
single throw (DPST)

-- two circuits simultaneously



single-pole,
double throw (SPDT)

-- one circuit, select
between two positions

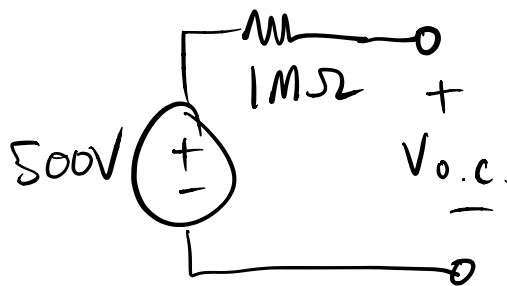


DPDT

↓ after two circuits, use numbers

3PDT, DT3T etc..

ex :



← no other components to complete circuit

∴ open-circuit voltage $V_{o.c.} = 500V$ because there's nothing to cause voltage division

∴ current source connected to open circuit is invalid

$$V = IR ; \text{ if } R = \infty, \quad \hookrightarrow \text{ i.e., unloaded } \quad V \rightarrow \infty$$

(unless $I = 0$)

Short circuit : equivalent to zero resistance ; perfect wire

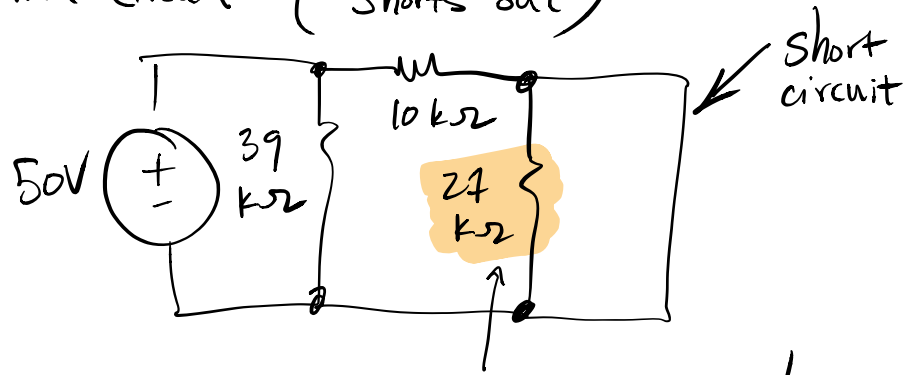
∴ wire sizes use AWG system (American Wire Gauge)

18 AWG wire : $\sim 6.83 \text{ m}\Omega/\text{foot}$ ← check this!

household wiring to breaker box : 12 : 14 AWG
 → smaller AWG # → larger wire, less resistance

- When a short is placed in parallel with any resistance, net resistance is zero; removes it from circuit ("shorts out")

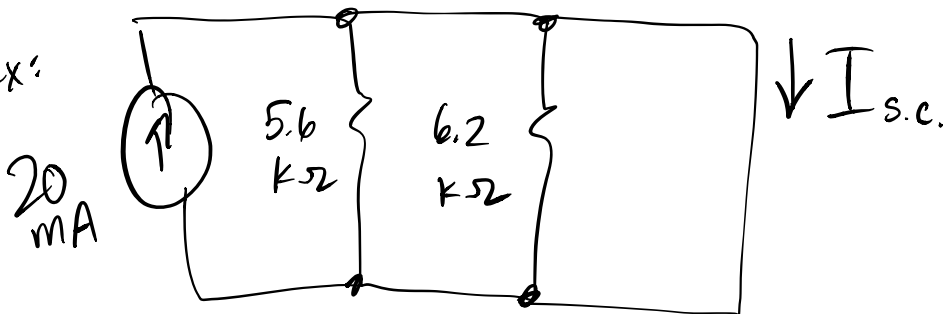
ex:



can be eliminated!

- a closed switch is a short circuit

ex:



- $I_{s.c.}$ is the short-circuit current

$I_{s.c.} = 20 \text{ mA}$ because no current flows through those resistors!

- "Shorted" voltage source $\rightarrow$ invalid circuit

$$I = \frac{V}{R} \quad \text{if } R \rightarrow 0, \text{ then } I \uparrow$$

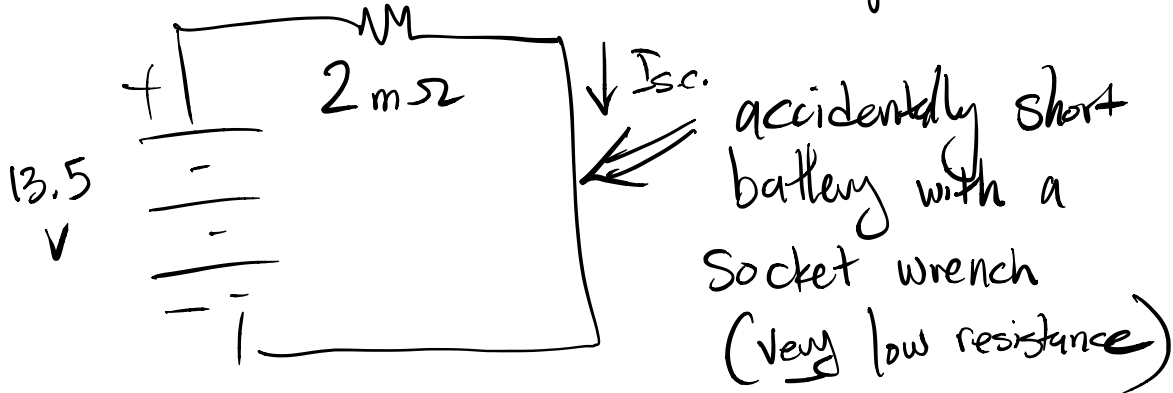
(unless $V = 0$)

ex: shorting a car battery



- .. practical car batteries do have some internal resistance
- .. even thick car battery wires have some resistance (#4 AWG or larger)

Still: low mΩ range



$$I_{s.c.} = \frac{13.5V}{2m\Omega} = 6.75 \text{ kA}$$

Vaporized metal, instantaneous heating of sulphuric acid, explosion, etc..

$$P_{2m\Omega} = I^2 R = (6.75 \times 10^3)^2 \cdot 2 \times 10^{-3}$$

$$91.13 \text{ kW} \leftarrow \text{Not for long!}$$

BE CAREFUL !!!

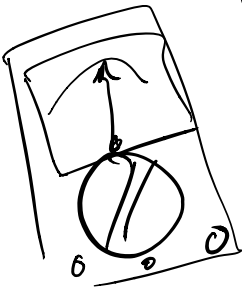
Measuring Voltage

Voltmeter: always placed in parallel w/ voltage to be measured

passive (requiring no external power): D'Arsonval galvanometer; needle-type meter, electromagnetic

-- very old concept, still used today

Simpson 260 (70+ years)



electronic: require external power (9V battery)
-- typical DMM

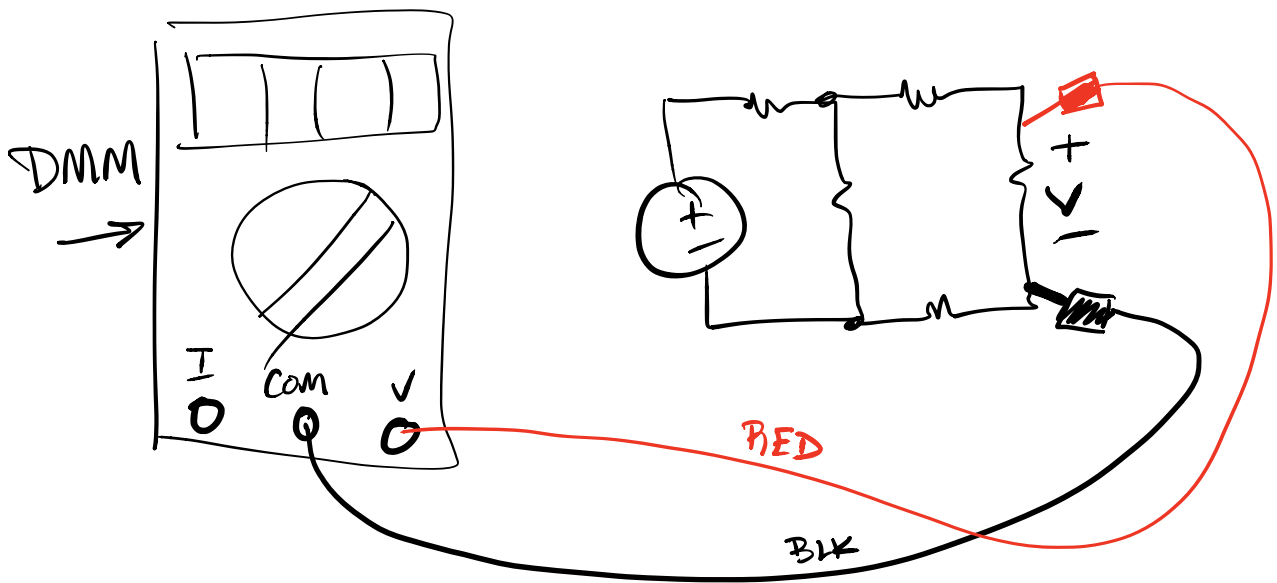
↳ digital multimeter

ideal voltmeter: looks like an infinite resistance to circuit under test

-- don't want to change circuit you're measuring

electronic voltmeters: often 1M Ω or better, good enough
passive voltmeters: not as good, higher voltage applications

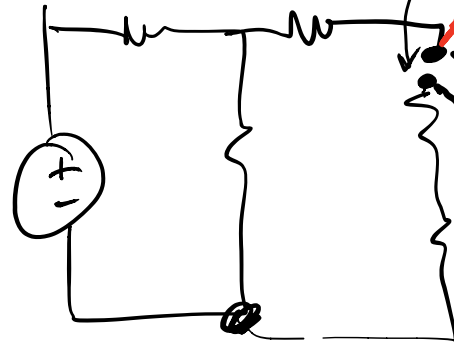
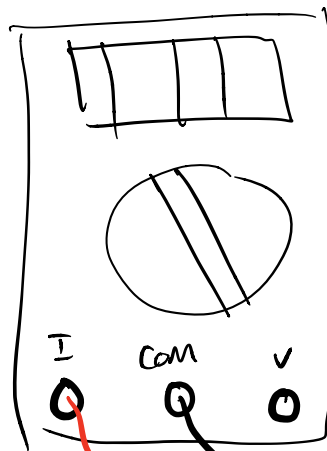
- .. must use correct port for voltage measurement



Measuring Current

- .. ammeter is always placed in series in a circuit to be measured
- .. circuit must be opened, and then completed with the ammeter (important!)
- .. ideal ammeter looks like a short circuit to the circuit under test (drops zero voltage, doesn't change anything)
- .. most electronic and D'Arsonval-type ammeters achieve a fraction of an ohm

.. again: use correct port!



break the circuit

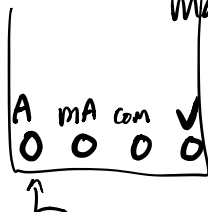
.. note convention of current measurement!
.. "into the I"

.. common mistake: forget that you're set up to measure current; ammeter is short circuit. then absent-mindedly measure a source voltage

short $\rightarrow$ huge current $\rightarrow$ blow fuse

.. keep spare fuses handy for your ammeter!

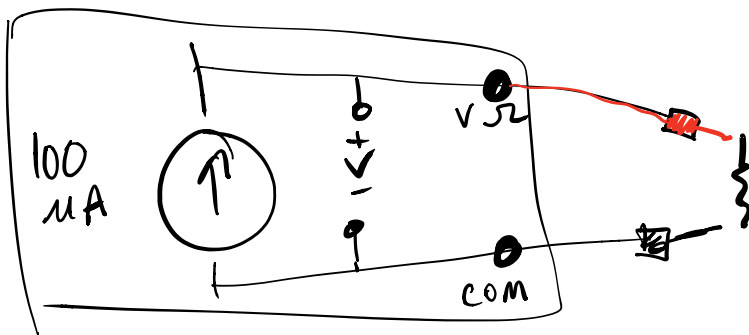
.. many ammeters have "high-current port" with extra low resistance and high-value fuse
.. careful!



- D'Arsonval galvanometer is really a microammeter converted into a voltmeter w/ series resistor or shunt for higher currents

Measuring Resistance

- most common method in modern DMM:
Use a precision test current source
($100\mu A$, $1mA$, etc.) and
measure voltage drop across resistor;
then resistance is proportional by Ohm's Law



then measure V internally

- don't attempt to measure resistance when circuit is energized!